Part 4

Successful AI Powered Apps



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PUSH PAST THE FIRST DRAFT

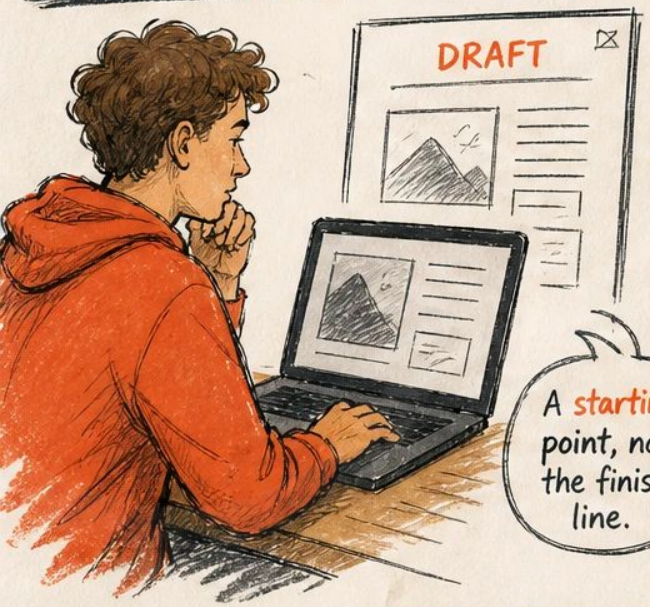
The goal is not getting AI to make something.



→ YOU DIRECT:

- Ask better questions
- Provide context & examples
- Define the outcome

FIRST OUTPUT:
A DRAFT, NOT A DELIVERABLE



A starting point, not the finish line.

CRITIQUE
IMPROVE
ITERATE



What's working?



What's unclear?



What's missing?



How can it be stronger?

Iteration is where the magic happens.

BETTER OUTPUT:
STRONGER IMPACT





REAL CHOICES. BETTER RESULTS.

STORY BEATS OUTPUT

A dashboard needs a point.

Strong projects tell a clear story so people can notice, compare, decide, or understand.

MORE OUTPUT



- ✕ Too much
- ✕ Hard to follow
- ✕ Hard to decide

STRONG STORY

REVENUE IS GROWING — DRIVEN BY NEW CUSTOMERS



FOCUS: INVEST IN NEW CUSTOMER CHANNELS TO ACCELERATE GROWTH.



CLEAR STORY. BETTER OUTCOMES.



NOTICE
What matters most.



COMPARE
See differences that count.



DECIDE
Choose with confidence.



UNDERSTAND
Turn insight into action.

SO WHAT?

GOOD DESIGN DOES MORE THAN WORK. IT MATTERS.



WHY IT MATTERS



It drives **real results.**
Good UX builds trust, increases engagement and achieves goals.

WHO IT HELPS



Inclusive design **opens doors.**
It removes barriers and creates better experiences for everyone.

WHAT WE LEARN



Every **insight** makes the next experience smarter and simpler.

DESIGN WITH PURPOSE. IMPACT FOLLOWS.

GOOD DESIGN ANSWERS THE QUESTION. GREAT DESIGN EXPLAINS SO WHAT?

VIBE CODING SKILLS

The main prompts from class come packaged as skills you install once and reuse.



1 PRD ARCHITECT



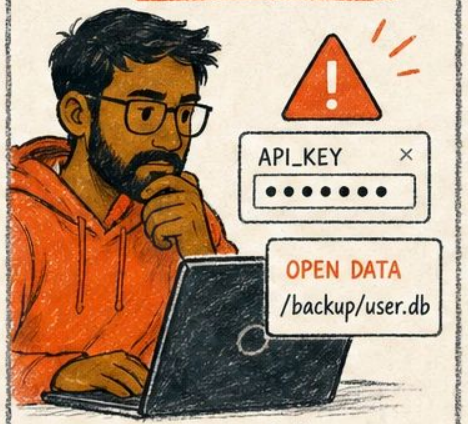
Plans your app.

2 USER TEST



Checks your app the way a new user would.

3 SECURITY CHECK



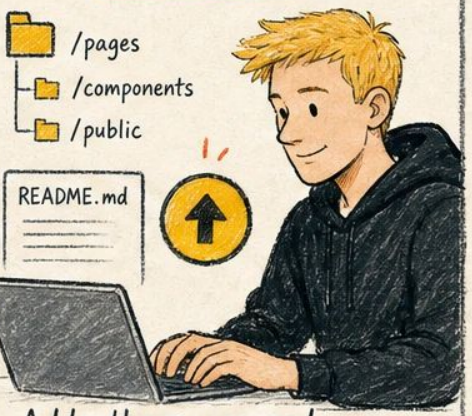
Finds exposed keys and open data.

4 PRIVACY CHECK



Compares your policy to your code.

5 SHIP FILES



Adds the pages and repo files every real site needs.

BUILD BETTER. SHIP CONFIDENTLY.

OPEN YOUR PROJECT IN A HARNESS

Let the AI read your whole codebase.



1 CONNECT TO GITHUB

Connect your app to GitHub first.

2 SAVE TO GITHUB

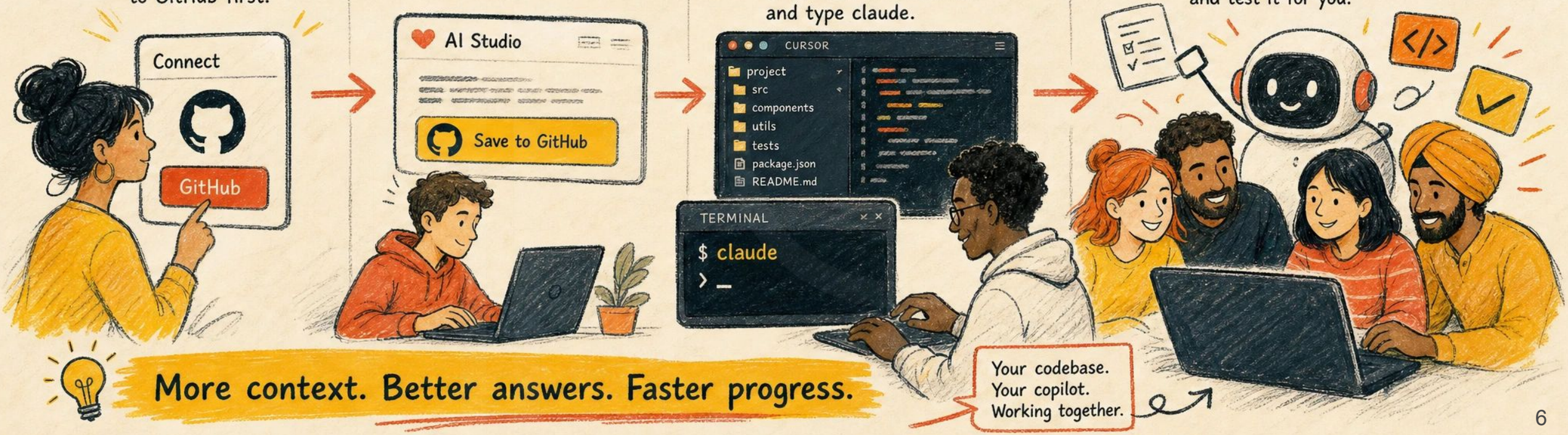
Lovable has a GitHub button and AI Studio can save to GitHub.

3 OPEN IN CURSOR OR TERMINAL

Open the folder in Cursor or clone it in a terminal and type claude.

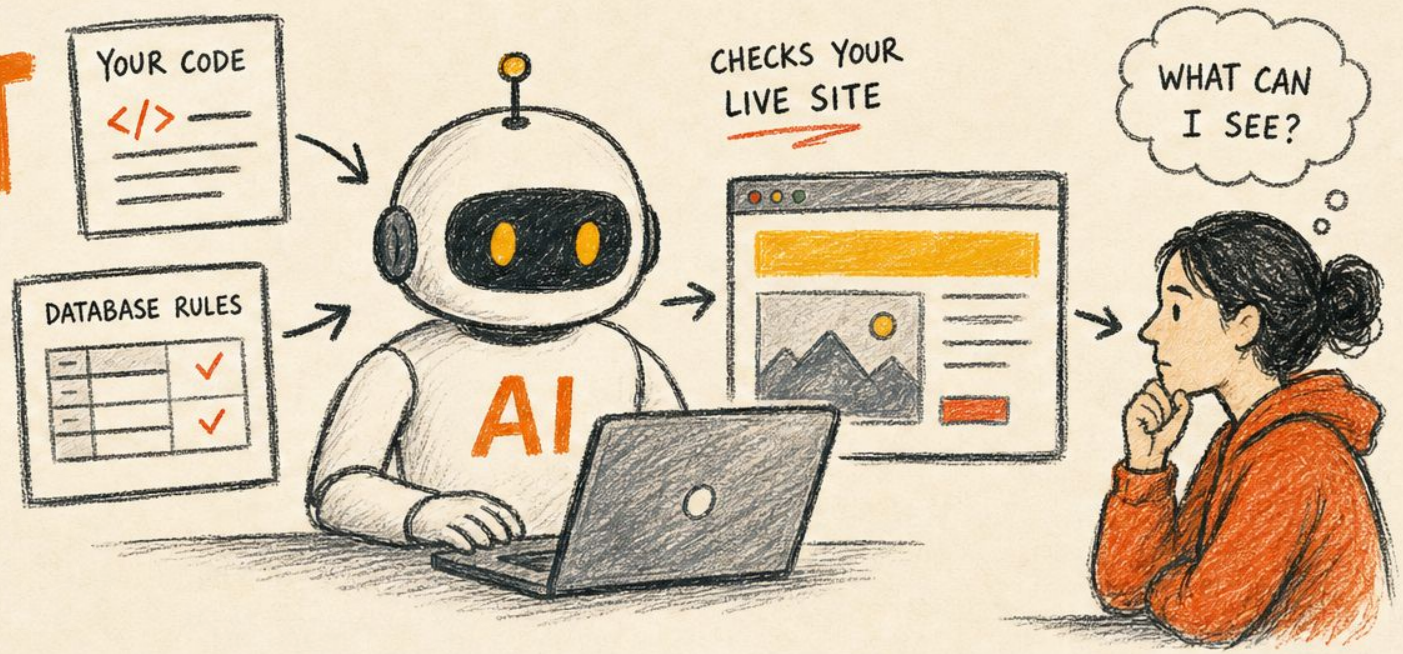
4 AI READS EVERYTHING. TESTS FOR YOU.

Now the AI can read every file and test it for you.



More context. Better answers. Faster progress.

WHAT THE SECURITY TEST CHECKS



An AI reads your code and database rules and then checks your live site the way a stranger would see it. **SIX CHECKS.**

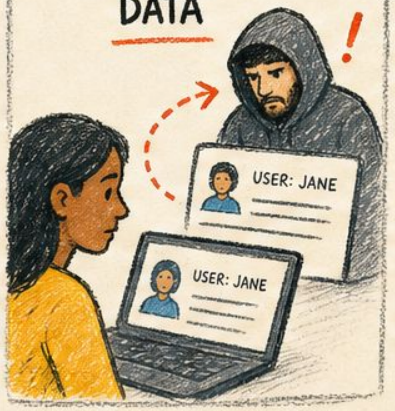
1 EXPOSED KEYS AND SECRETS



2 DATABASE RULES THAT LET ANYONE READ OR WRITE



3 ONE USER SEEING ANOTHER USER'S DATA



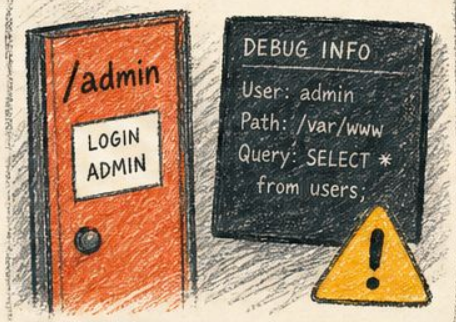
4 FORMS THAT TRUST WHAT PEOPLE TYPE



5 HTTPS ON EVERY PAGE



6 LEFTOVER ADMIN PAGES AND DEBUG INFO



SIX CHECKS. ONE GOAL: A SAFER SITE.

Service Costs

Every cloud service charges for what you use. If you plan to run a real site you need to understand what the meters measure.

HOSTING



Keeps your site online and serves pages to visitors.

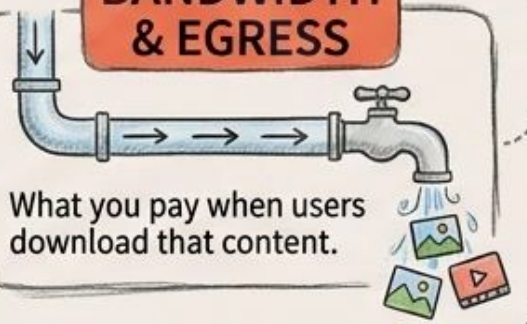
ALWAYS-ON SERVERS: Charge per hour.

FILE STORAGE



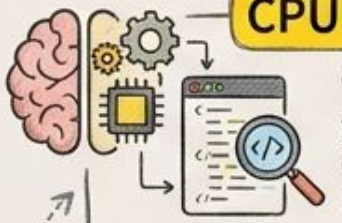
Holds your images, videos, and documents.

BANDWIDTH & EGRESS



What you pay when users download that content.

CPU / COMPUTE



What you pay when your server runs code to process a request.

SERVERLESS FUNCTIONS: Charge per call.

AI TOKENS



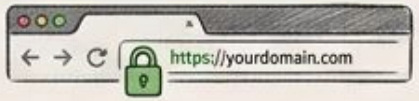
What you pay per word sent to or received from a model. Long prompts multiply fast.

DOMAIN NAME



Runs about $10 - $20 a year for a .com & auto-renews.

SSL CERTIFICATES



Formerly paid, now FREE through Let's Encrypt and included automatically on most platforms.

EMAIL

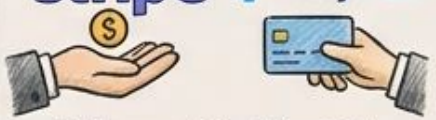


Branded address (hello@yourdomain.com) runs about $6/user/month via Google Workspace.

Resend SendGrid **TRANSACTIONAL EMAIL** Send signup confirmations & password resets. Free tiers usually cover the first few thousand/month.

PAYMENT PROCESSORS

stripe PayPal



Take roughly 3% + 30¢ per transaction.

SET BILLING ALERTS IN EVERY PLATFORM.

"Email me at ten dollars" beats finding out at four hundred.

No surprise bills!

ANALYTICS

You cannot improve what you do not measure.



DAU (Daily Active Users)

THE PULSE

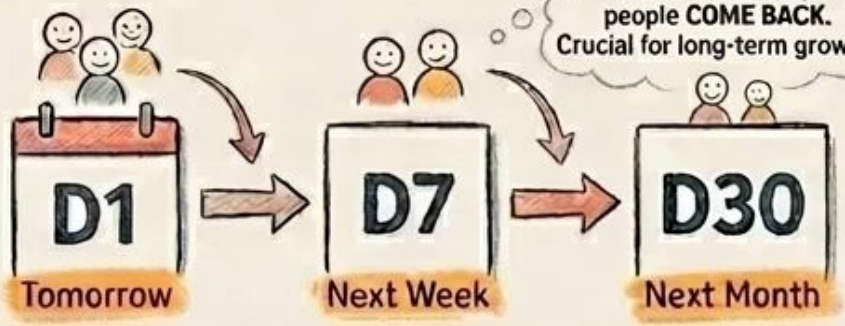
How many people use your site **TODAY**. A snapshot of daily engagement.

How many unique people people use your site daily.



RETENTION

Tracks how many people **COME BACK**. Crucial for long-term growth.



SHORTHAND: D1, D7, D30

ACTIVATION



IF ACTIVATION IS LOW, YOUR ONBOARDING IS BROKEN.

STANDARD TOOLS & OUT-OF-THE-BOX DASHBOARDS

Google Analytics 4 | AWS Analytics | Lovable Analytics | Google Cloud Analytics | Azure Analytics

SET A SMALL GOAL PER STAGE



Ten returning users next week beats building three new features nobody asked for.



Stages of a Site

Good sites are built in stages and each stage adds real infrastructure cost.

MVP

START FREE.
Start with the minimum viable product.

Gemini ChatGPT

Test the concept with free AI usage.

Local Storage

Remember Settings

Use Core Feature

Before asking for an account, let visitors use core features. Use browser local storage.

COSTS NOTHING TO HOST

Is anyone using this enough to justify the new cost?

STAGE TWO: NO-CODE BUILDER

Lovable
Use no-code builders.

Free Tier
yourapp.lovable.app
Subdomain (Limited)

Upgrade
Custom Domain (YourURL.com), More Credits Private Projects

~$25/MONTH

Is anyone using this enough to justify the new cost?

STAGE THREE: DATABASE & AUTHENTICATION

Authentication **Database**

Users save things across devices & need accounts with permissions.

Adds complexity and support time.

Is anyone using this enough to justify the new cost?

STAGE FOUR: SCALING (APIs, Storage, AI)

File Storage
Bills per gigabyte

APIs
Bills per call

Custom AI Features
Depending on requests

Payments

COSTS CLIMB FAST

Is anyone using this enough to justify the new cost?

INCOME

PRICING STRATEGY:
Price your product so that the income from your top tier covers the infrastructure cost of the free tier. If it does not, you are subsidizing strangers out of your own pocket.

Building Something People Want





Solve a real problem first.

Don't fall in love with your idea before someone else does.



Ship a rough MVP fast.

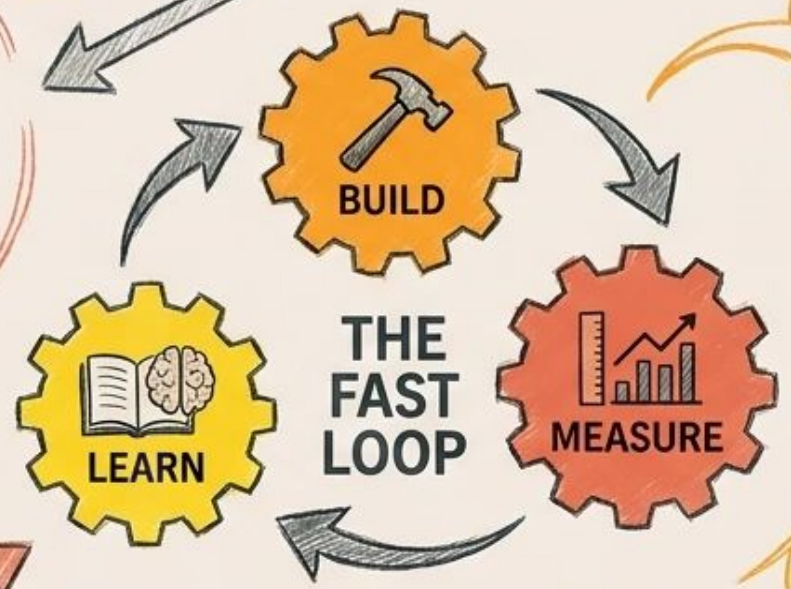
Your first version does not need every feature. It just needs to be good enough to put in front of real customers so you can start learning.

Talk to strangers who match your target customer, not your friends.



Talk to strangers who match your target customer, not your friends.

- Ask what hurts today
- What they use now
- What would make them switch
- What would make them tell a friend



Build, measure, and learn on a fast loop.
Small changes tested quickly beat big swings that take months.





Retention beats reach.
Aim for one hundred people who love your product, not a million who kind of like it.

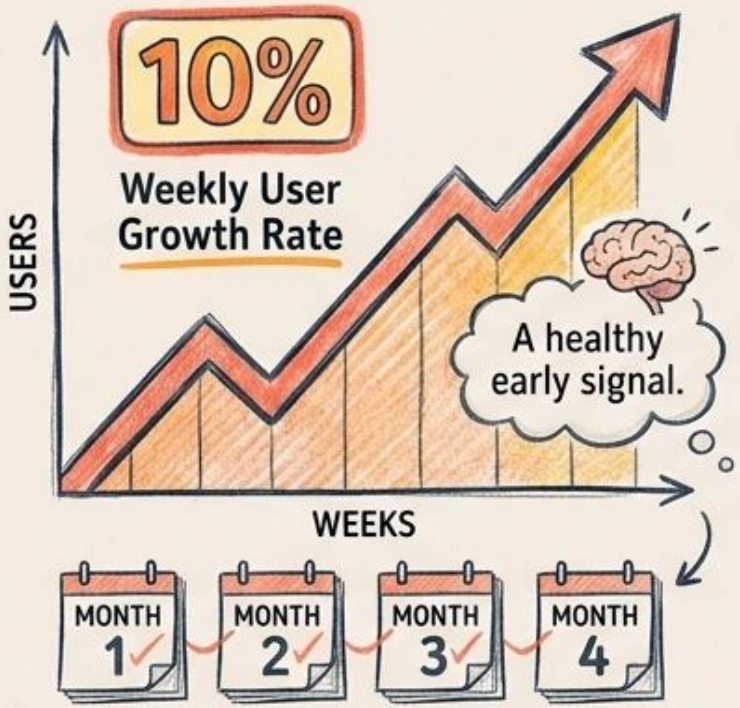


Product-market fit is when users spontaneously tell others about you.

Early Metrics That Signal PMF

A few numbers tell you whether your site has crossed from “some people use it” to “you have something real.”

Weekly User Growth Rate



Hit that for a few months in a row and you are on track.

“Very Disappointed” Score



If at least forty percent say “very disappointed” you have product-market fit.

Time-to-Value



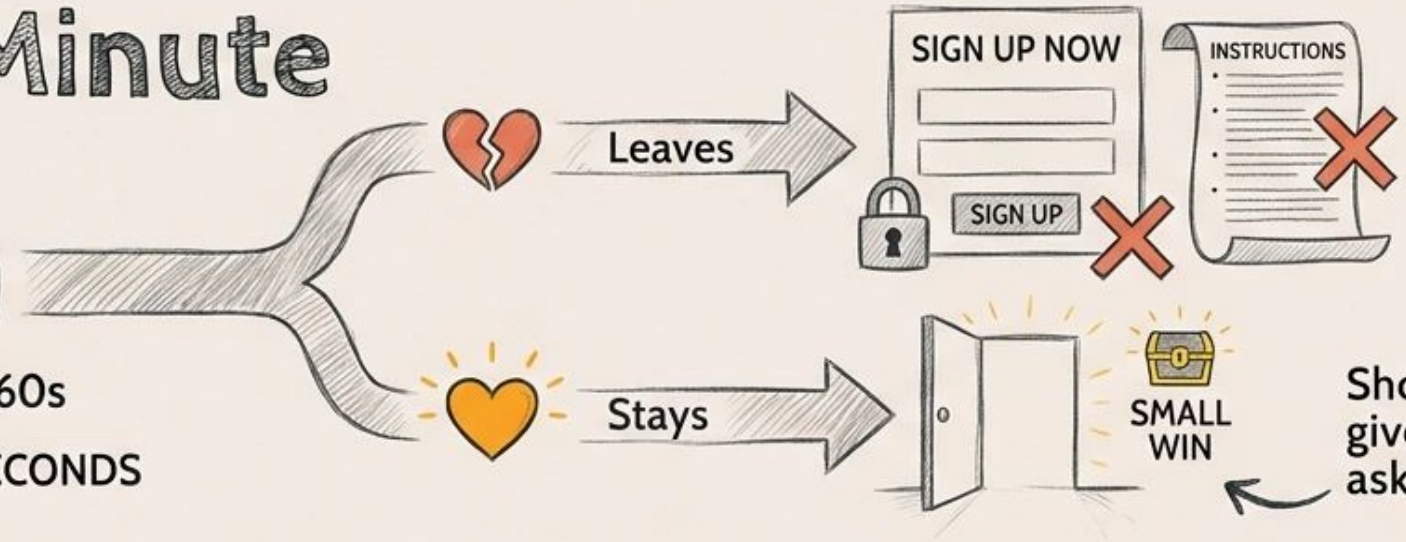
Shorter is better. A user who gets value in the first minute is hooked.

The First Minute



0s 60s

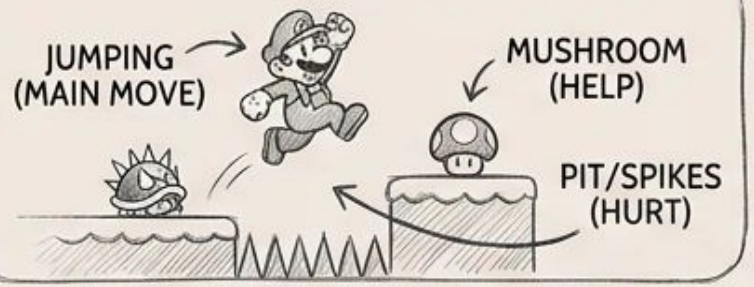
THE FIRST SIXTY SECONDS



Show meaningful value, give a small win *before* asking for anything.

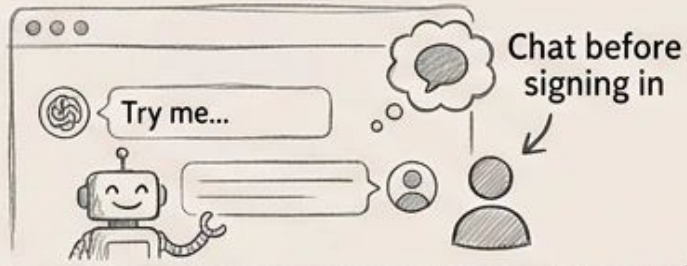
TEACH AS YOU PLAY

SUPER MARIO



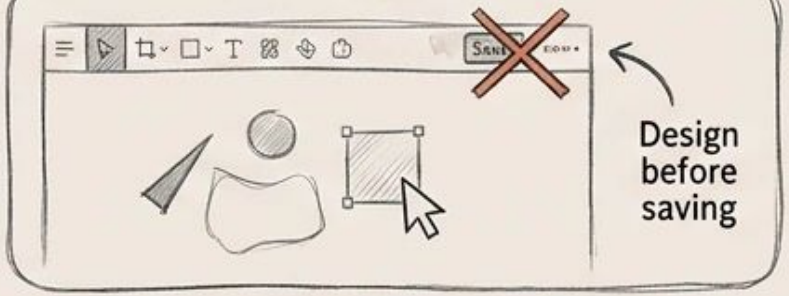
Zero text. First screen teaches mechanics instantly.

CHATGPT



Experience the core value immediately.

FIGMA



Start creating, lower the barrier.

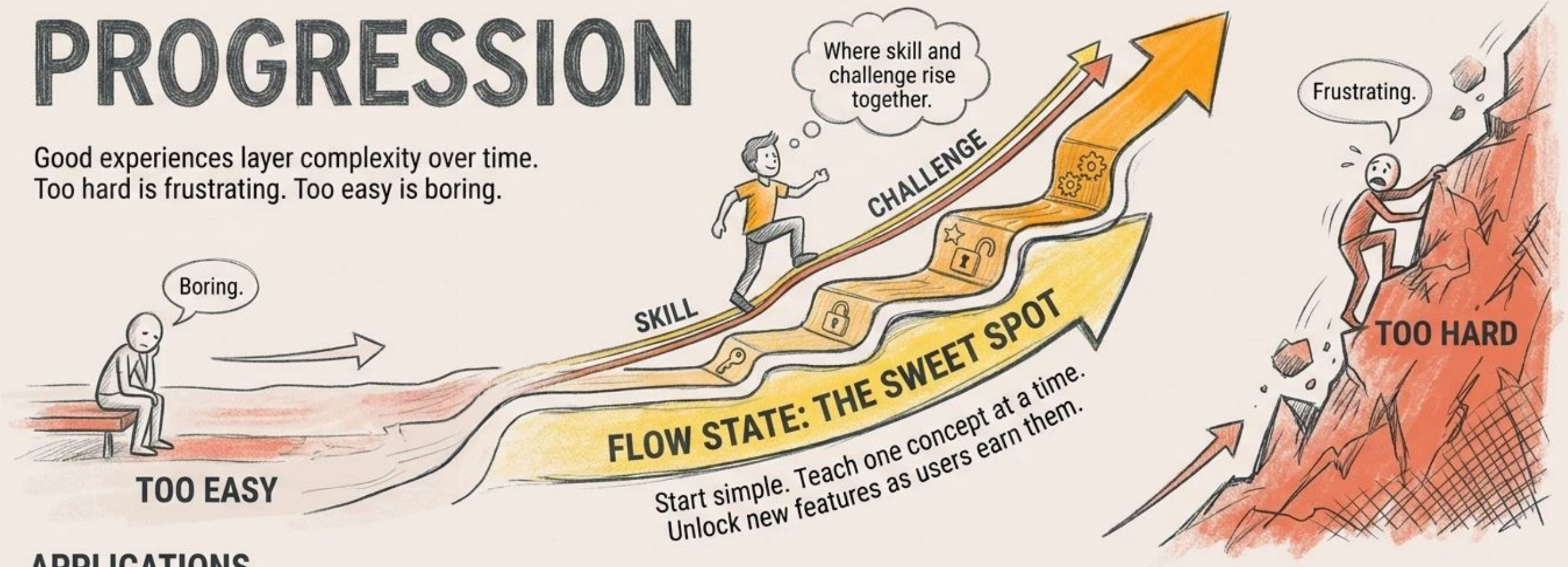




The best onboarding is the product doing something delightful, NOT a welcome modal.

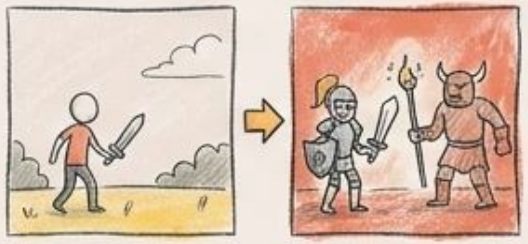
PROGRESSION

Good experiences layer complexity over time.
Too hard is frustrating. Too easy is boring.



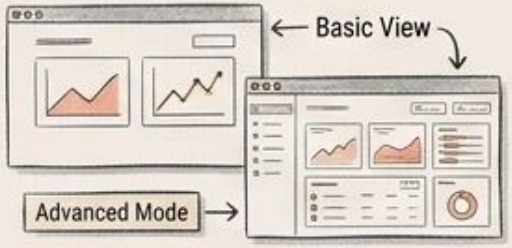
APPLICATIONS

GAMES



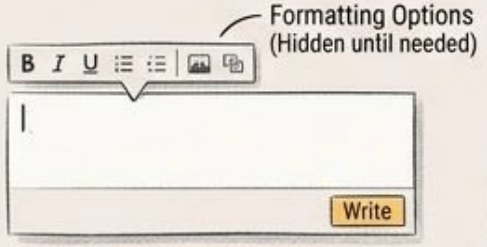
This applies to games and to any tool with multiple layers.

DASHBOARDS



Dashboards should bury power features behind an advanced mode.

EDITORS



Editors should hide formatting options until someone needs them.

CMS TOOLS



CMS tools should let new users publish a post before learning workflows. Show the user only what they need right now, and keep the door open for what's next.

Story and Stakes

Every product has a frame, even if you never write it down.
The frame is the answer to **"why am I here?"**

WITHOUT A FRAME



A budget app without a frame is a spreadsheet.



A fitness tracker without a frame is a list of numbers.



A writing app without a frame is a blank page.

WITH A FRAME



"You versus overspending" is a product people open every morning.



"You versus yesterday's you" is a product people wear on their wrist.

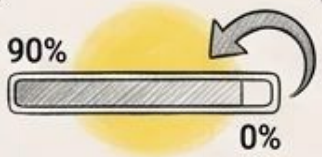


is a product people come back to.

STAKES ARE WHAT THE USER STANDS TO GAIN OR LOSE



A streak that breaks



progress that resets



an inbox that grows

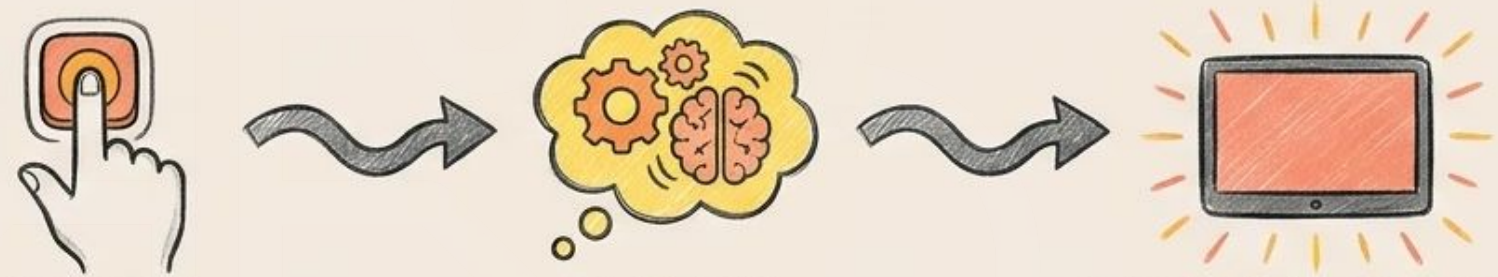


a deadline approaching

All of these feelings are **invented by the interface**, and all of them work.
Story does not mean **narrative**. It means giving the user an **emotional reason to return tomorrow**.

The clearer the frame, the less you need to convince anyone to use your product.

Feedback Loops



Every action a user takes should produce a visible response.

Silent Interfaces



Feel BROKEN even when they work.

No feedback = No connection.

Good Interfaces: Rewarding Actions

GAMES



Reward every tap, swipe, and kill with sound, color, animation, or points.

WEB APPS



Do the SAME. A save button that flashes green is more memorable than one that just changes its label.

Specific Examples

CLICKS ANIMATE



Clicks animate for tactile feel.

SAVES CONFIRM



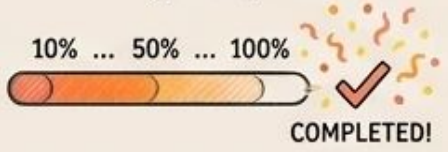
Saves confirm with a small success message.

ERRORS EXPLAIN



Errors explain what went wrong and what to do.

Small Touches, Big Impact





Toast messages on updates.



Small touches that tell the user the system HEARD them.

GAMIFICATION

You can borrow game mechanics without building a game.

PROGRESS BARS



Progress bars encourage completion because people finish what looks almost done.



STREAKS



Streaks encourage daily use because loss aversion beats reward seeking.



Duolingo, Snapchat, and Headspace all live on streaks.

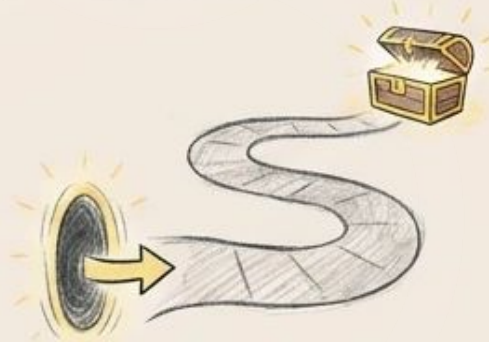
BADGES & LEVELS





Badges and levels turn status into a feature. Stack Overflow reputation, GitHub contributions, and Reddit karma all work this way.

ONE MORE TURN



The "one more turn" mechanic is what made Civilization famous. Players always feel they are one step from the next win, unlock, or breakthrough.



TikTok and infinite scroll work the same way, always one swipe from something interesting.

THE EXECUTION STRATEGY

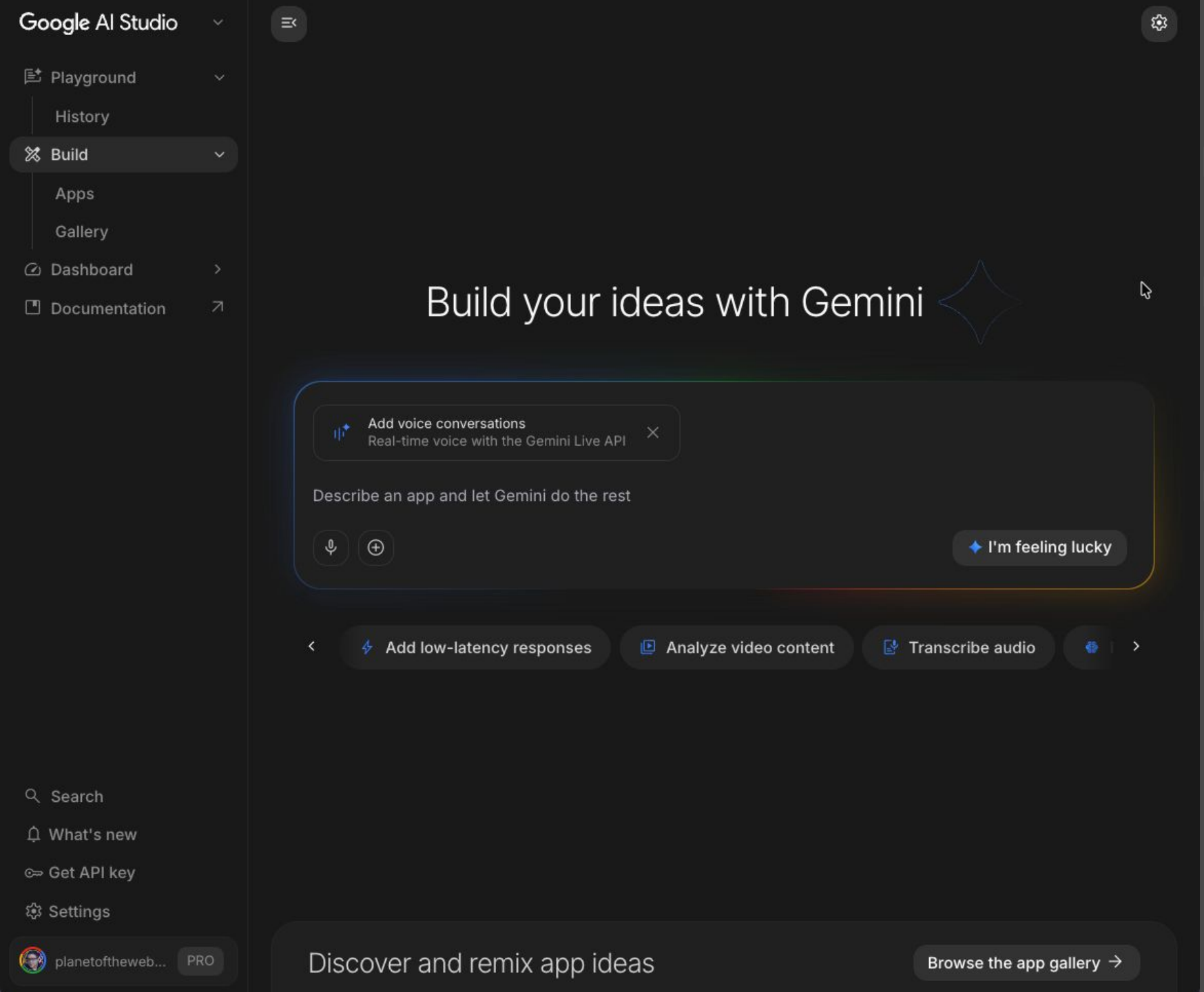
1. SHIP CORE EXPERIENCE FIRST → **2. ADD A STREAK** → **3. ADD A PROGRESS BAR**

Confirm people are using it.

Wait a month.

Layering mechanics before the product solves a real problem turns your site into a chore with no payoff.

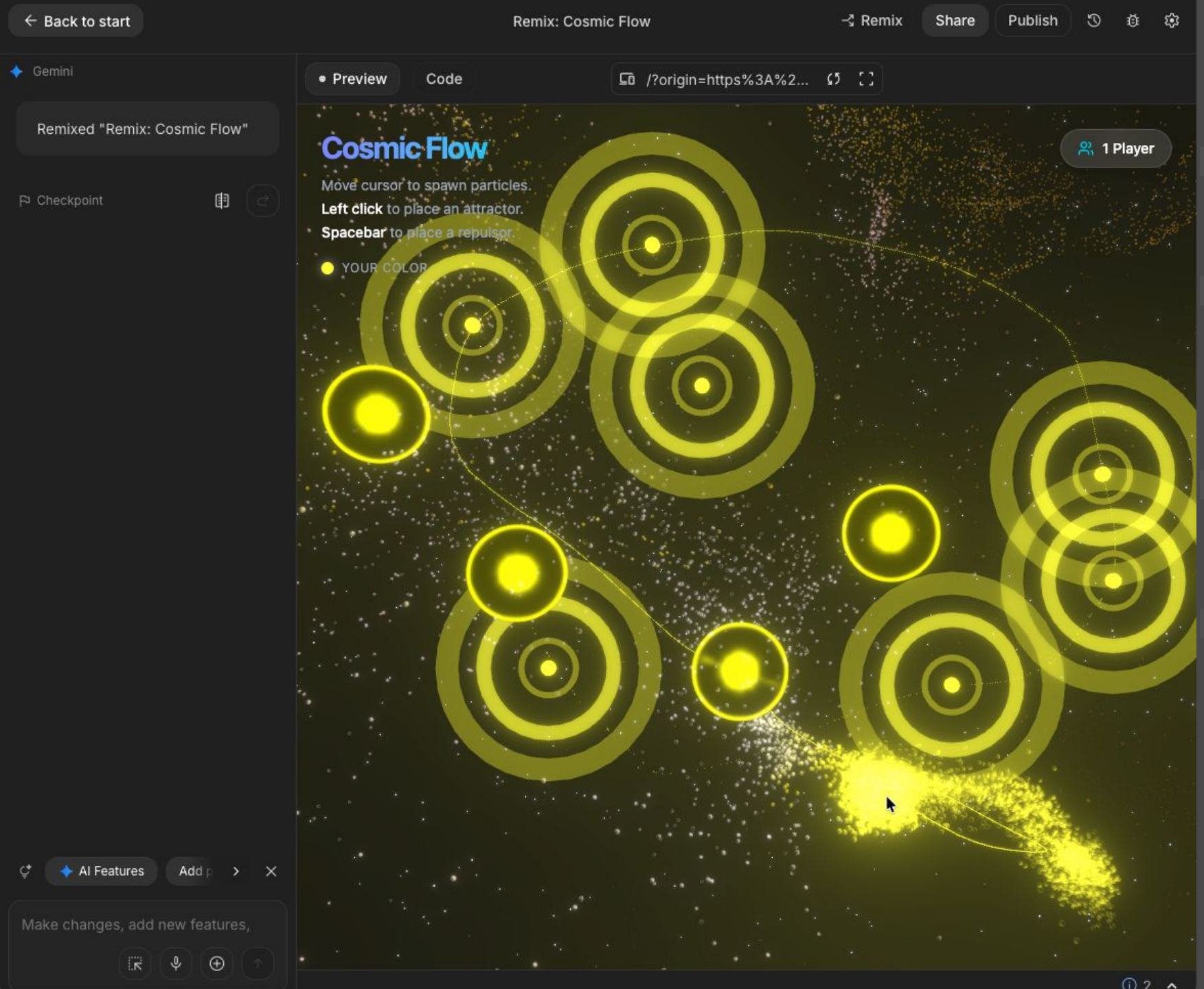
GAMIFICATION IS A LAYER ON TOP OF VALUE, NOT A SUBSTITUTE FOR IT.



Google AI Studio

Modern AI gives you a menu of free features to plug into any app. Google bundles them under Gemini in AI Studio and you get excellent free tiers and extra power with different Gemini accounts.

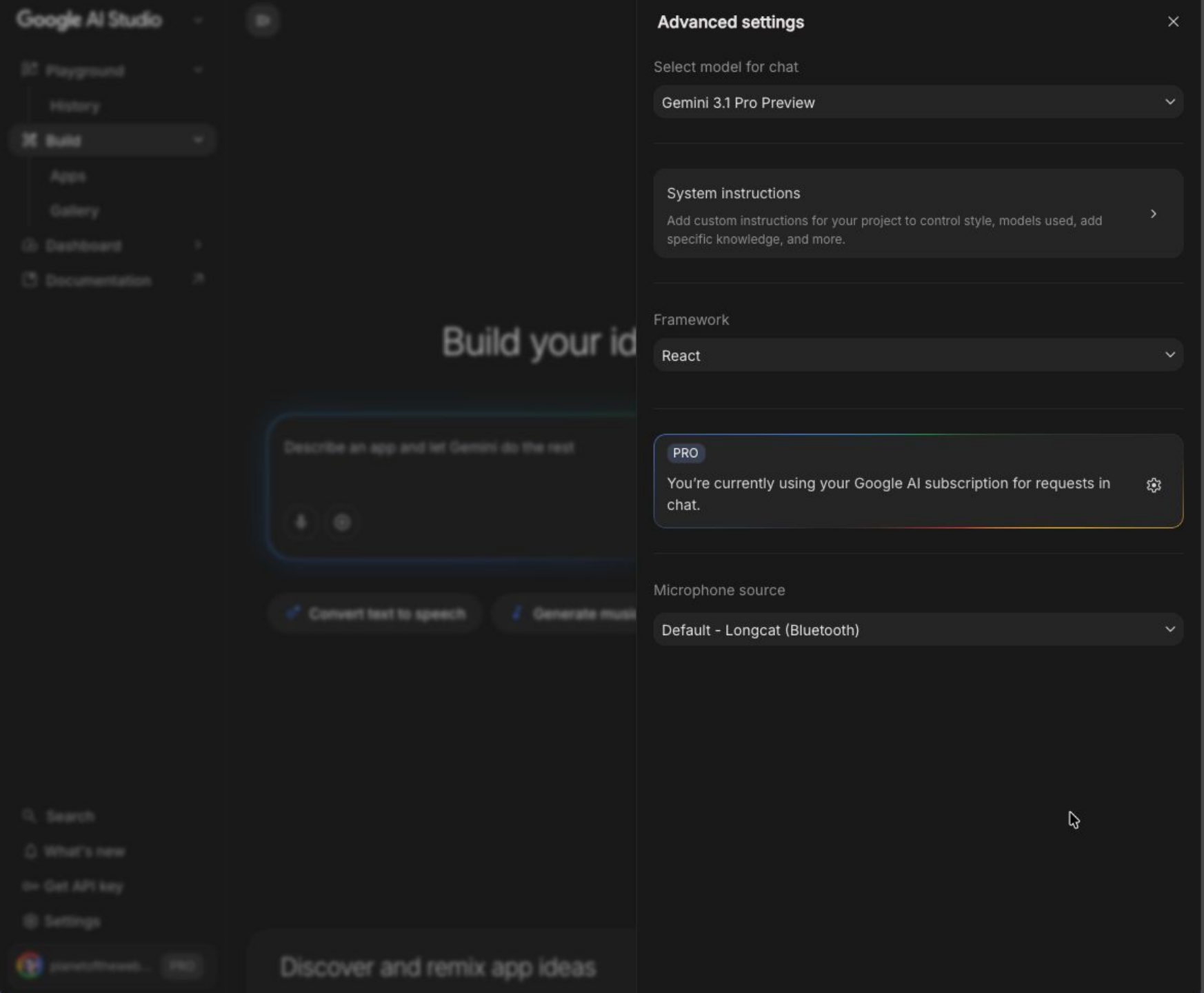
- **Text and reasoning:** chat, writing, summarization, Q&A, and function calling.
- **Vision:** describe images, read text in any language, and analyze diagrams or receipts.
- **Image generation:** create scenes, characters, and infographics with Nano Banana.
- **Video:** generate short clips with Veo, or summarize long clips with timestamps.
- **Audio:** text-to-speech voices, speech-to-text transcription, and generative music.
- **Live streaming:** two-way voice, webcam, and screen sharing through the Live API.
- **Data and auth:** Firestore database and Firebase authentication with Sign In With Google.
- **Google services:** real-time citations through Google Search grounding and live Google Maps embeds.



App Gallery

Unlike Lovable's gallery of full business apps, AI Studio's gallery is a showcase of what Gemini itself can do. Each sample is a small, remixable demo that highlights one capability.

- Click any sample to try it in the browser.
- Hit "Copy App" to fork the working code into your own workspace.
- Fastest way to see what is possible before you build from scratch.
    a. **Video analysis:** turn a YouTube clip into a quiz, flashcards, or a summary with timestamps.
    b. **Multimodal:** point a camera at something and get a real-time explanation.
    c. **Voice:** hold a natural conversation through the Live API.
    d. **Image generation:** Nano Banana playgrounds for scenes, characters, and layouts.
    e. **Maps and Search:** combine reasoning with live Google data.



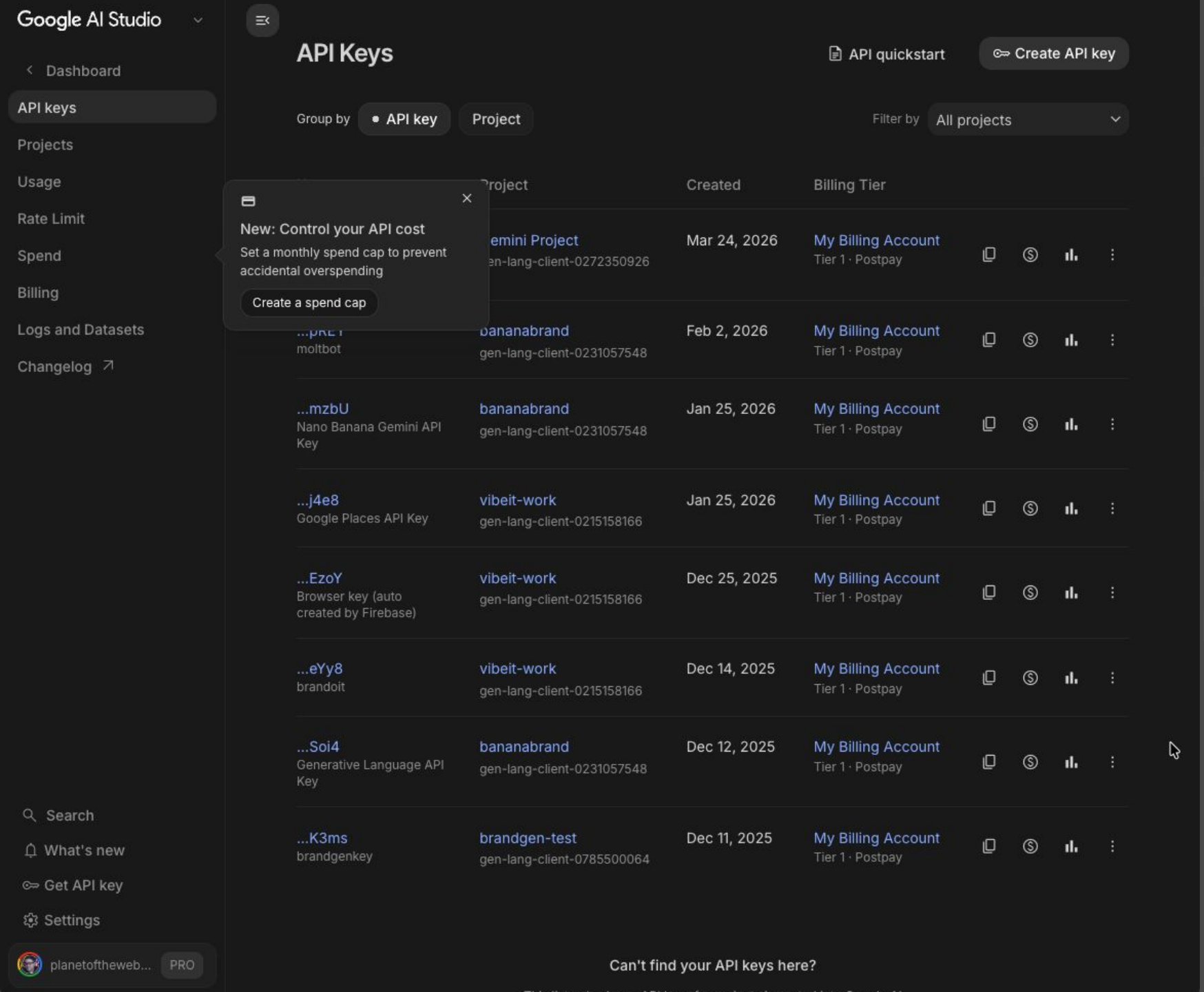
Settings

Click the settings gear next to the prompt box to fine-tune how your project gets built and run.

- **Select model for chat:** picks which Gemini variant (Pro, Flash, Flash-Lite) powers the chat inside your generated app.
- **System instructions:** a persistent prompt for style, tone, behavior, and specific knowledge the app should always apply.
- **Framework:** React by default. Sets the frontend stack Build uses for your generated code.
- **Subscription mode:** a Google AI Pro or Ultra subscription pulls requests from your paid tier instead of the free limits.
- **Microphone source:** picks the input device the Live API uses for voice capture during testing.

Writing good system instructions: don't over-specify. Pin down the parts you care about (fonts from fonts.google.com, brand colors, tone of voice, things to avoid) and let Gemini handle the rest. Over-prescribing fights the model and produces worse results.

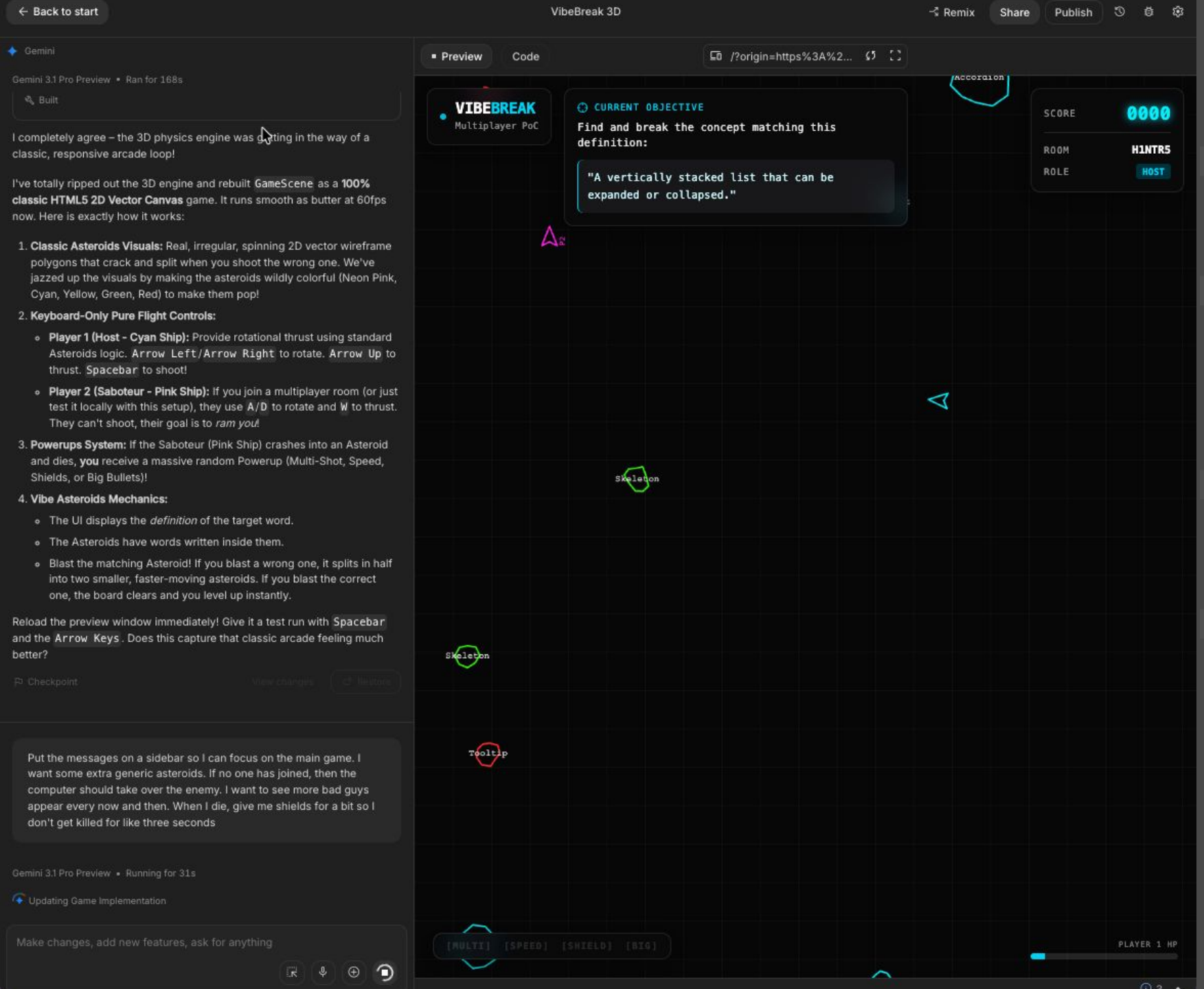
a.



API Keys

AI Studio handles your Gemini API key automatically, but you can manage keys manually from the sidebar.

- Your key attaches through a proxy when you hit Run in Build mode.
- Generated code shows `process.env.GEMINI_API_KEY` as a placeholder.
- When someone runs your shared app, AI Studio uses their key, not yours.
- Never paste a real key into the prompt, chat, or code.
- Manage keys under **Get API key** in the sidebar. Create, group by project, track usage per key.
- Set a **monthly spend cap** on each key to prevent accidental overspending.
- Billing tiers go from free (Tier 1 Postpay) up to higher limits as you scale.



Editor UI

Each Build project opens in a two-panel workspace. The left panel is your Gemini conversation and checkpoints. The right panel is the live app with preview, code, and publish controls.

1. **Prompt box:** type new instructions, ask follow-ups, attach files, or use voice to iterate on the build.
2. **Preview / Code toggle:** flip between the running app and the generated source.
3. **Preview URL and fullscreen:** copy the preview URL, refresh, or pop it into a full window.
4. **Remix:** fork the project into a new copy you own.
5. **Share:** create a link so others can try your app with their own key.
6. **Publish:** deploy to a real production URL on Google Cloud Run.
7. **History, debug, and settings:** version history, error logs, and the Advanced Settings panel.
8. **Status indicator:** shows active processes, errors, or warnings.

POPULAR GAMING FRAMEWORKS BUILT WITH

POWERING GAMES ACROSS 2D, 3D & WEB!



three.js

3D EXPERIENCES IN THE BROWSER



Create powerful 3D graphics and immersive worlds.

PHASER

2D GAMES MADE FUN & EASY



Build fast, engaging 2D games for web.

PIXI.JS

HIGH-PERFORMANCE 2D RENDERING

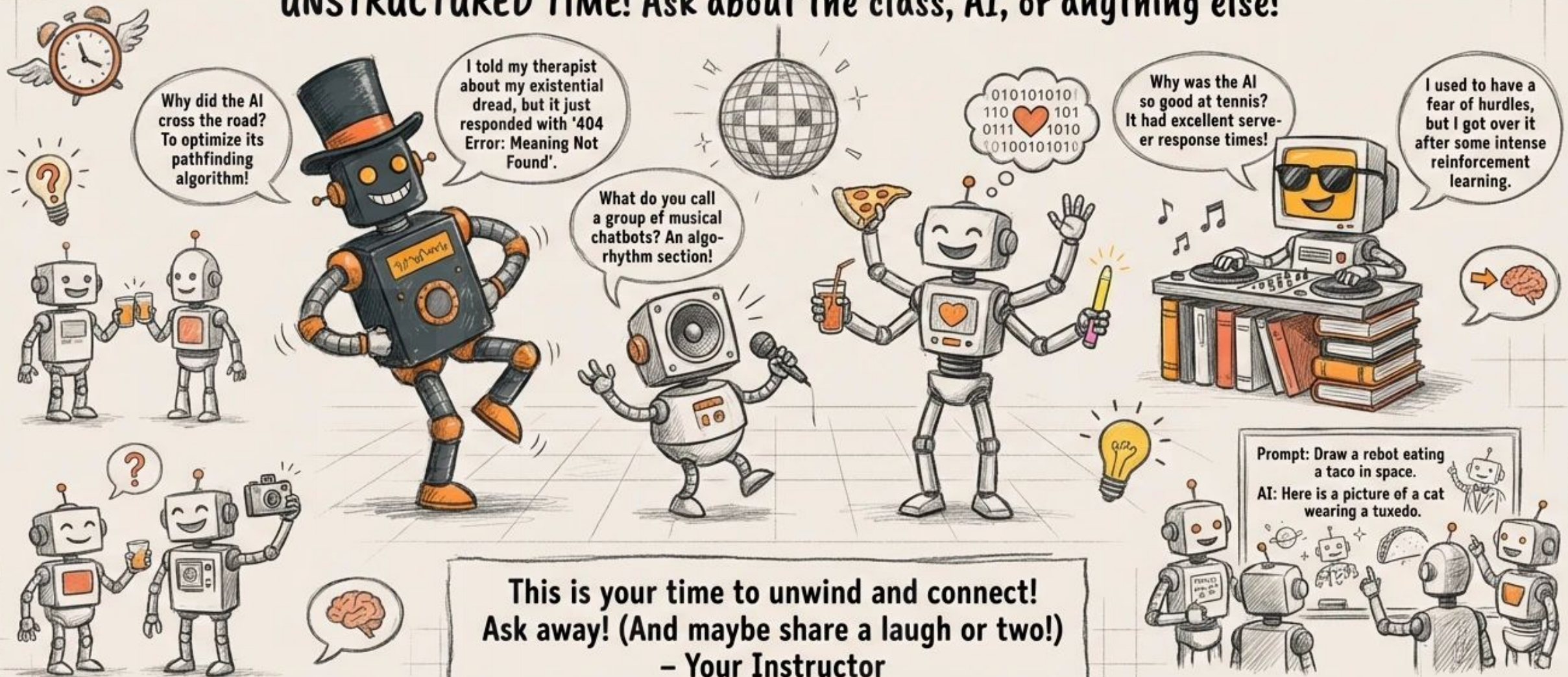


Stunning 2D visuals at high speed for modern games.

DIFFERENT STRENGTHS. ENDLESS POSSIBILITIES.

AMA (ASK ME ANYTHING) AFTERPARTY: THE CLASS CONCLUSION

UNSTRUCTURED TIME! Ask about the class, AI, or anything else!



This is your time to unwind and connect!
Ask away! (And maybe share a laugh or two!)
– Your Instructor